



Browse electrophysiology matrix

☐ Neurite Patterns

Neuron Type	V _{rest} (mV)	R _{in} (MΩ)	τ _m (ms)	V _{thresh} (mV)	Fast AHP (mV)	AP _{ampl} (mV)	AP _{width} (ms)	Max F.R. (Hz)	Slow AHP (mV)	Sag ratio
DG(36) DG Granule	-75.0	228.0	26.9	25.6	11.7	69.4	0.87	72.0	0.60	0.97
DG Adult-Born Immature Granule*	-45.6	2.9	67.4	-6.1	43.8					
DG Hilar Ectopic Granule				30.0	6.8	75.0		>68.0		0.99
DG Semilunar Granule	-78.0	128.3	19.6	37.0	21.0	85.6		>11.0		
DG Cajal-Retzius*										
DG Local-Projecting Cajal-Retzius*										
DG Mossy	-62.0	199.0	41.0	23.7	6.2	62.5	0.78	50.0	2.80	0.81
DG Mossy MOLDEN										
DG AIPRIM	-64.0	363.0	30.0	16.0	12.3	77.0	0.50	81.0	9.00	0.80
DG Axo-axonic	-65.1	73.9	7.7	13.0	7.0	78.0	0.42	85.0	4.50	
DG Granulosum-targeting Axo-axonic*										
DG Axo-axonic GRALDEN*										
DG Basket	-62.0	43.0	10.0	18.3	20.0	72.6	0.25	230.0	2.30	0.97
DG Basket GRALDEN*		103.0					0.14	268.0		
DG Basket CCK+										
DG Interneuron Specific COLTAR*	-60.8	812.0		22.9		75.7	0.91	30.3		
DG COM-MOPP*										
DG HICAP	-60.5	182.2	14.8		18.6	50.0	0.51	26.5	0.10	
DG Interneuron Specific IMOT*										
DG Hilus-Associated Interneuron*	-57.1	186.7	9.3				0.62	141.5		
DG Interneuron Specific HIMOLD*	-47.5	889.0		11.3		71.6	0.88	25.2		
DG HIPP		24.0	5.0	19.0	10.7	44.5	1.06			0.96
DG Recurrent HIPP*										
DG HIPP-CAP*										
DG Recurrent HIPP-CAP*										
DG Interneuron Specific HIPRO*	-47.5	889.0		11.3		71.6	0.88	25.2		
DG HIPPOM	-65.0	371.0	35.0	26.2	13.1	80.8	0.72	69.0	3.10	0.82
DG Ivy*		354.0	21.6		16.1	54.7	1.50			
DG MOCAP		480.0	51.4		20.0	72.8	1.70		0.00	0.88
DG MOLAS	-54.5	198.2	18.4	15.2	11.0	43.8	0.86	50.0	3.50	
DG Molecular Layer*										
DG Outer Molecular Layer	-57.0	252.3	16.9					13.5		
DG Total Molecular Layer	-54.5	198.2	18.4	15.2	11.0	43.8	0.86	50.0	4.50	
DG MOPP	-75.6	147.0	6.8	45.6	14.6	41.7			2.19	0.95
DG Neurogliaform	-75.6	147.0	6.8	45.6	14.6	41.7			2.19	0.95
DG Wide-Arbor Neurogliaform*										
CA3(34) CA3 Pyramidal	-60.5	126.0	61.0	28.0	10.2	97.6	1.00	40.0	7.50	1.01
CA3 Athorny Pyramidal*	-68.5	124.6		18.2	2.0	76.8				
CA3c Pyramidal	-62.0	52.0	16.0	15.0	3.5	66.0			5.10	
CA3c Umbrella Pyramidal*	-62.0	52.0	16.0			67.4	2.74			
CA3 Giant	-57.0	595.0	67.0	22.0	14.0	76.0	1.10	>50.0	7.10	0.68
CA3 Granule	-78.0	139.0	17.1	27.4	8.0	104.3	0.95	>100.0	1.75	0.99
CA3 Axo-axonic	-58.0	182.8	8.9	10.5	29.3	80.5	0.69		9.00	
CA3 Horizontal Axo-axonic				26.0	4.5	61.0			5.60	0.75

Legend:

* indicates new v2.0 neuron type

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Values are selected from the source with the greatest number of measurements taken at preferred experimental conditions (rats, patch clamp, & body temperature when available).

Hover over a value for standard deviation, experimental conditions, number of measurements, and number of sources.

Pale versions of the colors in the matrix indicate interpretations of neuronal property information that have not yet been fully verified.

	CA1 Ivy	-71.0	72.8	7.6	30.1	13.6	44.0	0.80	3.00	0.98
	CA1 Interneuron Specific LM-R									
	CA1 Interneuron Specific LMO-O									
	CA1 LMR	-53.1	352.0	32.9	36.9	19.5	86.9	0.80	0.20	0.92
	CA1 LMR-LM*									
	CA1 Interneuron Specific LMR-O*									
	CA1 LMR-R*		220.0	42.6				1.10	52.0	
	CA1 Interneuron Specific LMR-R									
	CA1 LMR Projecting									
	CA1 Neurogliaform	-63.1	215.3	12.4	32.4	20.4	73.0	0.90	>60.0	9.00
	CA1 Neurogliaform Projecting	-63.1	215.3	12.4	44.4	20.4	46.5	0.90	52.8	9.00
	CA1 O-LM				32.3	20.4	81.8	0.88		4.00
	CA1 Recurrent O-LM	-75.0	70.0	12.8	25.0	16.1	58.8	0.60	>150.0	5.00
	CA1 O-LMR	-55.7	190.0	39.4	34.0	19.4	84.4			5.00
	CA1 Recurrent O-LMR*									0.68
	CA1 Interneuron Specific O-R									
	CA1 O-RP*	-66.0	364.0	38.0	26.0	21.4	63.0	0.44	53.2	
	CA1 Interneuron Specific O-Targeting									
	QuadD									
	CA1 Oriens/Alveus	-51.0	42.0	5.6	9.4	8.1	43.0	1.10		6.00
	CA1 Oriens-Bistratified	-66.0	364.0	38.0	26.0	21.4	63.0			0.85
	CA1 Oriens-QuadA Projecting									
	CA1 OR-LM	-68.0	225.0	68.0	15.1	8.4	86.0			4.60
	CA1 O-RPO*									0.71
	CA1 Perforant Path-Associated	-55.4	84.2	11.9	9.3	7.0	69.4	0.73		12.40
	CA1 Perforant Path-Associated QuadD	-64.0	216.0	46.0	27.0	22.0	61.0	0.77	>40.0	1.30
	CA1 QuadD-RP*									0.79
	CA1 Quadrilaminar	-58.5			21.1	11.9	57.7			6.90
	CA1 Interneuron Specific R-O									
	CA1 R-Receiving Apical-Targeting	-57.0	116.0	13.0	25.0	25.0	48.0	0.54		3.30
	CA1 R-RO*		220.0	42.6				1.10	52.0	0.84
	CA1 R-RP*									
	CA1 Radiatum	-55.8	96.3	16.2	19.4	16.3	68.4			4.20
	CA1 Interneuron Specific RO-O									0.88
	CA1 Interneuron Specific RO-R*									
	CA1 RO-RP*									
	CA1 RP-RPO*									
	CA1 Schaffer Collateral-Associated	-60.6	482.6	26.1	13.5	21.0	73.0	1.06	>95.0	5.50
	CA1 Schaffer Collateral-Correlated*									0.49
	CA1 Schaffer Collateral-Receiving R-Targeting									
	CA1 SO-SO	-59.0	401.0	38.0	24.0	15.0	48.0	1.12	>160.0	4.80
	CA1 SO-SO Projecting*	-59.5	196.4		16.8	14.6	65.3	0.70		0.69
	CA1 Trilaminar	-64.0	216.0	46.0	27.0	22.0	61.0	0.77	>130.0	4.70
	CA1 Radial Trilaminar	-57.0	116.0	13.0	27.0	25.0	48.0	0.54	>145.0	0.82
Sub(6)	Sub CA1-Projecting Pyramidal	-60.0	150.0		11.0	8.0	41.0			0.89
	Sub EC-Projecting Pyramidal				18.0		92.0			2.00
	Sub Axo-axonic									
	Sub Pyramidal Multipolar Interneuron*	-67.0	137.0			23.0	64.0	0.34		
	Sub Wide-arbor Polymorphic*	-49.9	387.8				91.8	1.50	26.0	
	Sub Recurrent Polymorphic-Molecular*	-49.9	387.8				91.8	1.50	26.0	
EC(36)	EC LI-II Multipolar-Pyramidal	-63.0	430.0	25.0	15.6	5.5	77.5	1.22	>50.0	4.70
	EC LI-II Pyramidal-Fan	-56.5	400.0	16.0	17.8	11.5	69.0	1.25	>90.0	3.00

MEC LII Oblique Pyramidal	-60.5	400.0	15.8	8.0	7.2	69.0	1.25	>40.0	3.00	0.66
MEC LII Stellate	-60.5	30.2	8.9	8.0	7.2	69.0	1.25	>40.0	3.00	0.79
MEC LII-III Pyramidal-Multiform	-63.0	430.0	25.0	15.6	5.5	77.5	1.22	>25.0	4.70	0.78
EC LII-III Pyramidal-Tripolar	-60.5	400.0	16.0	8.0	7.2	69.0	1.25	>40.0	3.00	0.66
LEC LIII Multipolar Principal	-66.5	450.0	29.0	23.6	10.6	70.0	1.38	>25.0	4.06	0.90
MEC LIII Multipolar Principal	-61.0	468.0	43.5	19.5	12.7	76.0	1.45			0.86
EC LIII Small Pyramidal	-61.0	468.0	43.5	19.5	12.7	76.0	1.45			0.86
LEC LIII Complex Pyramidal	-66.5	450.0	29.0	23.6	10.6	70.0	1.38	>25.0	4.06	0.90
MEC LIII Complex Pyramidal	-61.0	468.0	43.5	19.5	12.7	76.0	1.45			0.86
MEC LIII Bipolar Complex Pyramidal	-61.0	468.0	44.0	17.2	8.2	76.0	1.45			0.86
EC LIII Pyramidal	-66.5	450.0	29.0	23.6	10.6	70.0	1.38	95.0	4.06	0.90
EC LIII Stellate	-61.0	468.0	43.5	19.5	12.7	76.0	1.45			0.86
EC LIII-V Bipolar Pyramidal	-68.0	490.0	36.0	20.0	7.2	70.0	1.48	>29.0	4.50	0.86
EC LIV-V Pyramidal-Horizontal										
EC LIV-VI Deep Multipolar Principal	-59.8	272.3	26.6	14.5	5.7	80.4	1.39	>70.0	1.20	0.95
MEC LV Multipolar-Pyramidal	-66.5	605.0	37.0	28.7	7.8	61.0	1.80		0.00	0.86
MEC LV Principal*										
MEC LV Pyramidal	-61.8	605.0	37.0	10.0	11.6	61.0	1.80		0.00	0.86
EC LV Deep Pyramidal	-66.5	605.0	37.0	28.7	7.8	61.0	1.80	>70.0	5.00	0.86
MEC LV Superficial Pyramidal	-66.5	605.0	37.0	28.7	7.8	61.0	1.80		0.00	0.86
MEC LV-VI Pyramidal-Polymorphic	-63.0	480.0	41.0	27.7	8.1	73.0	2.10	>33.0	1.60	0.91
LEC LVI Multipolar-Pyramidal	-64.0	450.0	26.0	27.9	7.4	74.0	1.70	>10.0	1.20	0.90
MEC LI-II Polar Interneuron*										
EC LII Axo-axonic										
MEC LII Basket										0.99
EC LII Basket-Neurogliaform	-75.5	206.0	11.3	32.0	4.8	64.2	0.99			
EC LIII Bipolar Interneuron*	-61.1	418.0	38.5	19.8	11.7	67.3	1.25			
LEC LIII Multipolar Interneuron										
MEC LIII Multipolar Interneuron	-54.0	382.0		27.0	38.0	87.8	0.40		3.50	0.80
MEC LIII Superficial Multipolar Interneuron	-64.0	143.0		22.0	31.0	81.2	0.20		4.30	0.95
MEC LI Neurogliaform*	-65.1		7.5							
EC LII-III Superficial Polymorphic Interneuron*	-74.1	385.0	34.6	22.1		77.4	0.93			
EC LIII Pyramidal-Looking Interneuron	-71.4	50.5	24.5	29.9	10.8	58.9			9.80	
MEC LIII Superficial Trilayered Interneuron	-62.0	36.8	12.7	8.0	11.2	66.8			1.10	0.89